

Modal Logic: Tableaux Examples

2026-02-18

Example 11: $\Diamond A \rightarrow \Diamond\Diamond A$ in KT

Example 11: $\Diamond A \rightarrow \Diamond\Diamond A$ in KT

1. $\mathbb{F} 1, \Diamond A \rightarrow \Diamond\Diamond A$ ✓ Assumption
2. $\mathbb{T} 1, \Diamond A \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond\Diamond A \rightarrow \mathbb{F} 1$

Assume the formula is false at world 1. A false conditional gives a true antecedent ($\Diamond A$ true at 1) and a false consequent ($\Diamond\Diamond A$ false at 1).

- | | | |
|----|--|-------------------------|
| 1. | $\models 1, \Diamond A \rightarrow \Diamond\Diamond A$ ✓ | Assumption |
| 2. | $\top 1, \Diamond A$ ✓ | $\rightarrow \models 1$ |
| 3. | $\models 1, \Diamond\Diamond A$ | $\rightarrow \models 1$ |
| 4. | $\top 1.1, A$ | $\Diamond \top 2$ |

True $\Diamond$ (line 2) introduces world 1.1: since $\Diamond A$ is true at world 1, there must be an accessible world where A holds. We call it 1.1.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond A \rightarrow \Diamond\Diamond A$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond A$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond\Diamond A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, A$ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{F} 1.1, \Diamond A$ | $\Diamond \mathbb{F} 3$ |

False $\Diamond$ (line 3) is a reusable rule: since $\Diamond\Diamond A$ is false at world 1, $\Diamond A$ must be false at every world accessible from 1. World 1.1 is accessible from 1, so $\Diamond A$ is false at 1.1.

1.	$\mathbb{F} 1, \Diamond A \rightarrow \Diamond\Diamond A$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond A$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond\Diamond A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, A$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{F} 1.1, \Diamond A$ ✓	$\Diamond \mathbb{F} 3$
6.	$\mathbb{F} 1.1, A$	$\mathbb{T} \Diamond 5$
	⊗	

The $\mathbb{T}\Diamond$ rule: since $\Diamond A$ is false at world 1.1, and in KT every world accesses itself, A must be false at 1.1 itself. But line 4 says A is true at 1.1 — contradiction. The branch closes. This formula is **valid** in KT.